Foundamentals

1) Voltage, V is always a defined difference

$V_{element}$ = V(+ terminal of element) - V(- terminal of element)

2) Voltages labelled at nodes [Node voltage] always has its node as the +ve terminal

V_{node} = V(+) - V(-)

= V(node) - V(reference node) most of the time ground

= V(node) - 0

2) Polarity of elements of unknown voltage can be chosen arbitrarily

You can always choose a reference direction for a variable voltage, V

If your assumed polarity is opposite to reality, the calc. voltage will be negative.

3) Current direc. can be chosen arbitrarily for an unknown current

You can always choose a reference direction for a variable current, I

If your assumed direc. is opposite to reality, the calc. current will be negative.

4) $V_{element}$ = IR only works under passive sign convention (current enters + terminal), otherwise V = -IR (current enter - terminal)

5) KVL sign depends on how you cross the polarity, not current.

Crossing + to - give -V, lose potential

Crossing - to + give +V, gain potential

} Normal convention but can be vice versa.

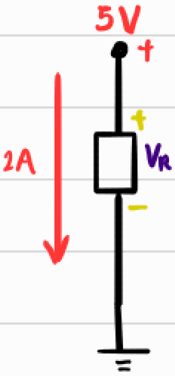
[Just be consistent]

Example using rule 1, 2, 3, 4

chosen arbitrarily | given

But not 2A cuz its not a variable

We can choose the polarity of V_R arbitrarily



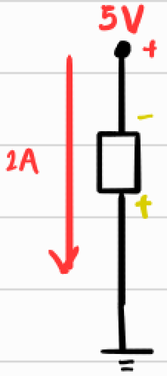
$V_R = V(+) - V(-)$
$= 5 - 0$
$= 5$

Since current in +

$V_R = IR$

$5 = 2R$

$R = 2.5\ \Omega$



$V_R = V(+) - V(-)$
$= 0 - 5$
$= -5\ V$

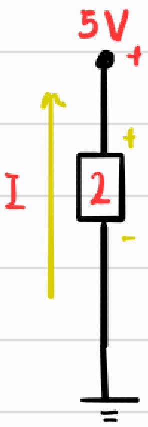
Since current in -

$V_R = -IR$

$-5 = -2R$

$R = 2.5\ \Omega$

We can choose direc of I & polarity of V_R arbitrarily



$V_R = V(+) - V(-)$
$= 5 - 0$
$= 5$

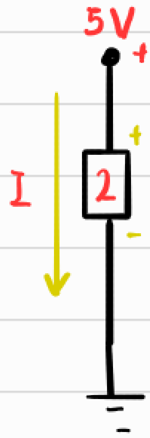
Since current in -

$V_R = -IR$

$5 = -I(2)$

$I = -2.5\ V$

$= 2.5$ downwards



$V_R = V(+) - V(-)$
$= 5 - 0$
$= 5$

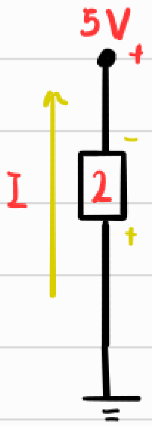
Since current in +

$V_R = IR$

$5 = I(2)$

$I = 2.5\ V$

$= 2.5$ downwards



$V_R = V(+) - V(-)$
$= 0 - 5$
$= -5$

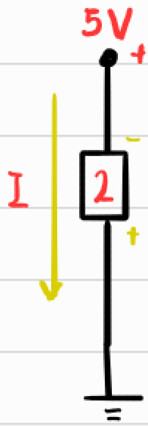
Since current in +

$V_R = IR$

$-5 = I(2)$

$I = -2.5\ V$

$= 2.5$ downwards



$V_R = V(+) - V(-)$
$= 0 - 5$
$= -5$

Since current in +

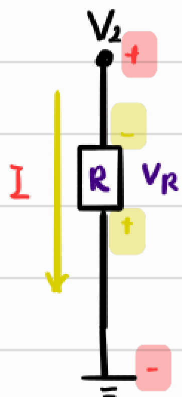
$V_R = -IR$

$-5 = -I(2)$

$I = 2.5\ V$

$= 2.5$ downwards

Extra Example



$V_R = V(+) - V(-)$

$V_R = 0 - V_2$

$V_R = -V_2$

$= -(IR)$

$V_R = -IR$ — ①

Take Note:
$V_2 \neq -IR$
because I flows
from V_2^+ to V_2^-

$V_R = -IR$ — ②

Take Note:
I flows from
V_R^- to V_R^+